

Artificial Intelligence

Section 1

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Section Content

- Introduction to AI
- Subfields of Artificial Intelligence
- AI Libraries In Python
- AI Simple Application Examples
- Python Review

• What is AI

- ❑ According to the father of Artificial Intelligence, John McCarthy, it is “*The science and engineering of making intelligent machines, especially intelligent computer programs*”.
- ❑ “*It is a branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions.*”
- ❑ Intelligence can include skills such as learning, reasoning, and solving problems, language understanding, etc.

• AI Applications

- Speech Recognition
- Machine Translation
- Facial Recognition and Automatic Tagging
- Virtual Personal Assistants
- Self Driving Car
- Planning

Introduction to
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Python Review

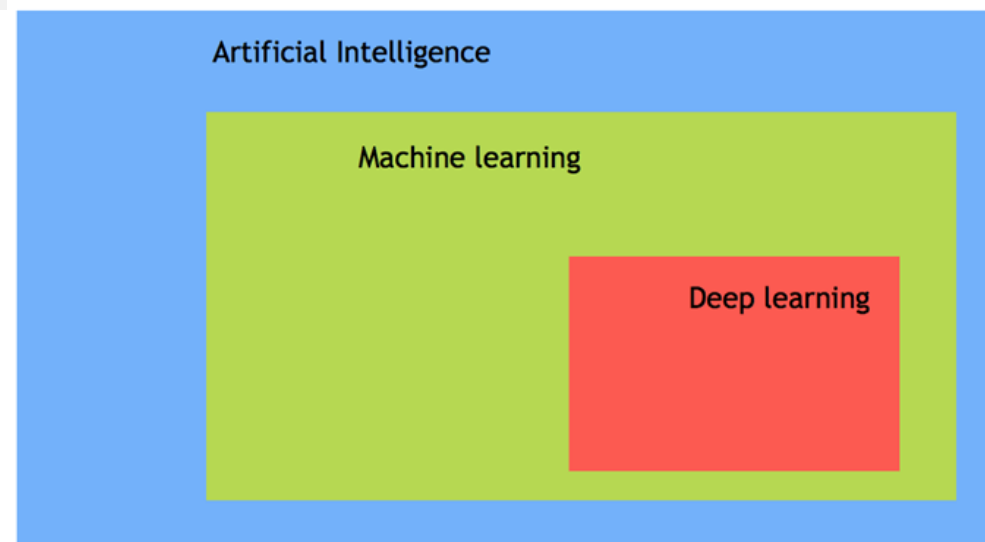
○ Here, are some important subfields of Artificial Intelligence:

➤ Machine Learning:

- Machine learning is an application of AI that enables systems to learn and improve from experience without being explicitly programmed. Machine learning focuses on developing computer programs that can access data and use it to learn for themselves.

➤ Deep Learning:

- Deep learning is a sub-field of machine learning. Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is inspired by the human brain.



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➤ Natural Language Processing:

- Natural Language Processing (NLP) is a field of artificial intelligence (AI) that enables computers to analyze and understand human language, both written and spoken.

➤ Computer Vision:

- Computer vision (CV) is the subcategory of artificial intelligence (AI) that focuses on building and using digital systems to process, analyze and interpret visual data.

➤ Expert Systems:

- An expert system is an interactive and reliable computer-based decision-making system that uses facts and heuristics to solve complex decision-making problems. It is also considered at the highest level of human intelligence. The main goal of an expert system is to solve the most complex issues in a specific domain.

➤ Fuzzy Logic

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- Machine Learning

- scikit-learn

- Built using the popular NumPy, SciPy and matplotlib libraries, scikit-learn is a Python AI library that makes implementing machine learning algorithms easier. Its extensive library of functionality ensures you don't have to reinvent the wheel if you want to add machine learning to your project.

- TensorFlow

- TensorFlow is a framework, created by Google, to make highly complex machine and deep learning algorithms accessible to the masses. The computations are written in Python but executed in C++, which is much faster than Python. TensorFlow saves time and increases program speed.

- XGBoost

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- Natural Language Processing

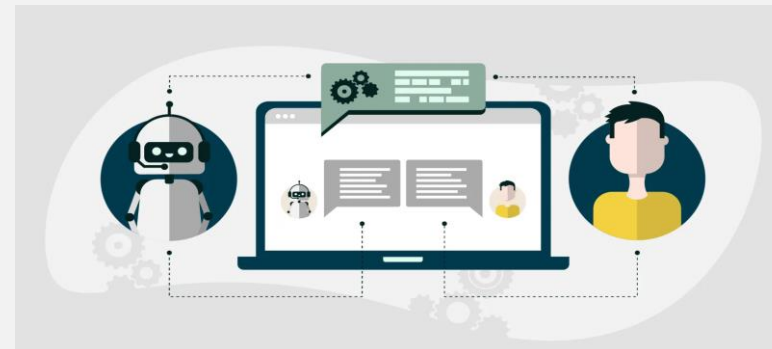
- Some examples are chatbots, writing analytics tools (such as Grammarly) and smart assistants in your phones and homes.

- NLTK

- NLTK stands for Natural Language Toolkit. It's a Python AI library that makes trivial linguistics simple through a variety of defined functions and interfaces. NLTK is a general-purpose NLP library (or 'toolkit') that belongs in any language-based project.

- spaCy

- GensimS



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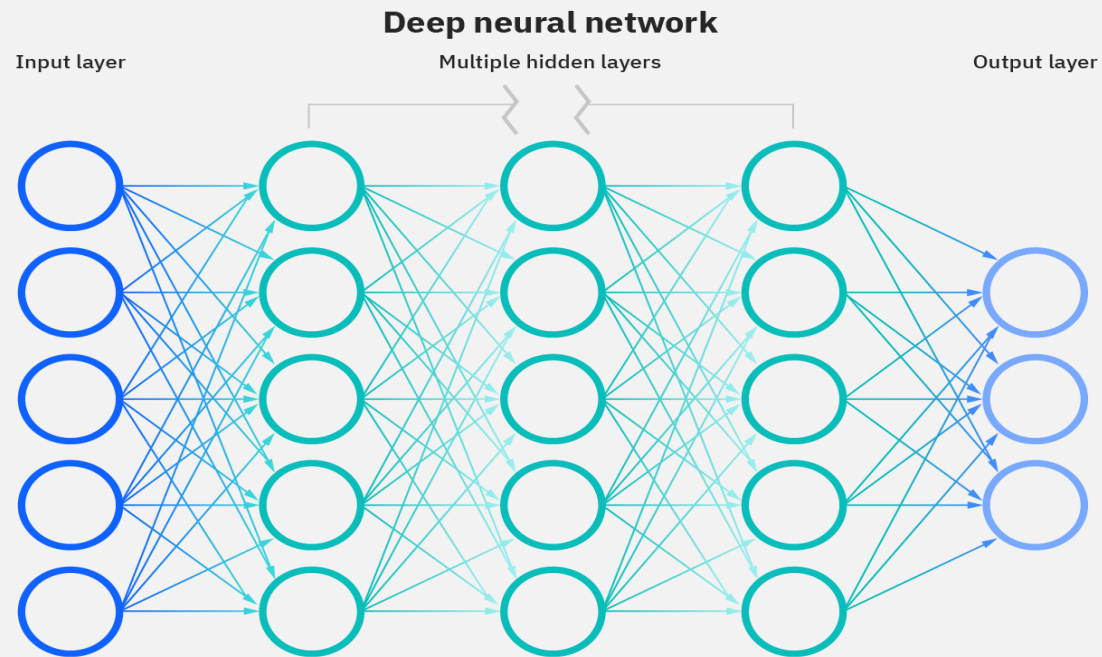
- Neural Networks

- Neural networks can allow us to implement facial recognition algorithms, predict the weather more effectively or even simulate marketing campaigns for businesses.

- FANN

- ffnet

- PyTorch



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- Computer Vision

- OpenCV
- SimpleCV



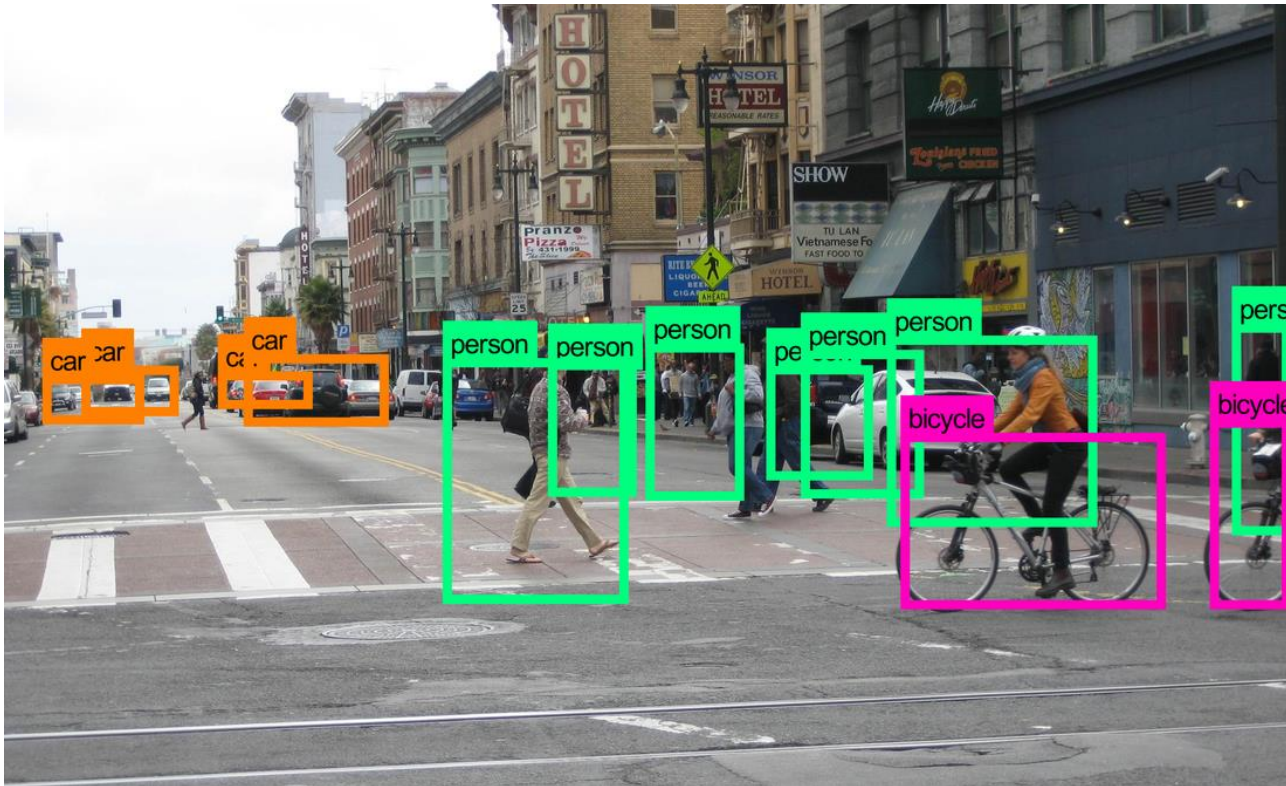
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- Expert Systems

- Want the advice of an expert? You can ask one (for a price) or you can get a machine to think like an expert for you. As their name suggests, expert systems aim to model an expert's reasoning process using a knowledge base (heaped with relevant data), an inference engine (to reason based on the data), and an explanation interface (to communicate its reasoning to the user).

- PyCLIPS

- Experta

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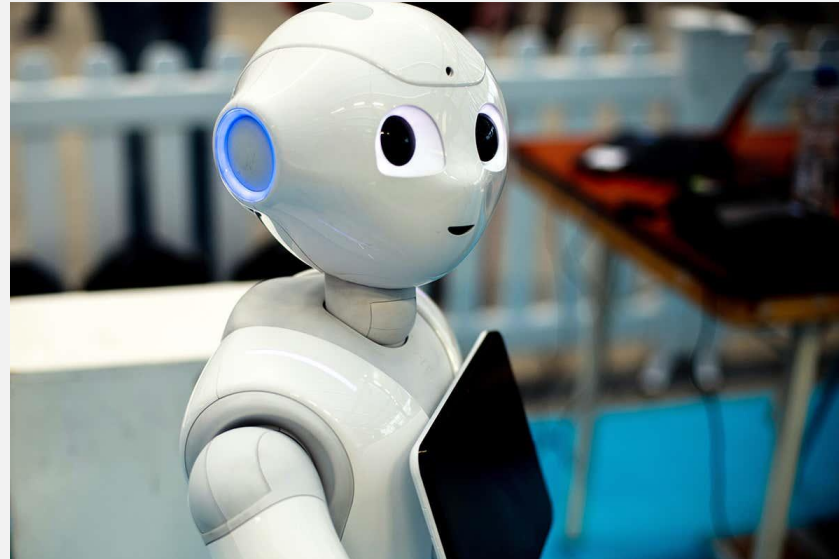
- Robotics and Autonomous Vehicles

- Most artificial intelligence is usually applied to software, not hardware. With robotics, AI meets the physical world through a plethora of different sensors modeling the human senses. From cameras to model sight and radar to model physical touch, AI algorithms must interpret the data from these sensors to allow a machine to function safely and effectively.

- AirSim

- Carla

- Bullet



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- ChatGPT
- DALL·E 2
- Zoo Animal Classification
- Stop Sign Object Detection
- Text Summarization

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Python Review

- Comment
- Variables
- Basic Operators
- Console Output / Input
- If Statement
- While/ For
- Lists / Tuples / Dictionary
- Functions

- Declaration
- Python has five standard data types –
 - Numbers
 - String
 - List
 - Tuple
 - Dictionary
- Using
- Naming
- Random

Comments

Variables

Basic Operators

Console O/I

If Statement

While / For

Lists / Tuples
/Dictionary

Functions

Arithmetic Operators and Comparison Operators and Logical Operators

Addition	+
Subtraction	-
Multiplication	*
Division	/
Modulus	%
Exponent	**
Floor Division	//
==	
!=	
>	
<	
>=	
<=	
and	
or	
not	

Comments

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Syntax

```
if expression1:  
    statement(s)  
elif expression2:  
    statement(s)  
elif expression3:  
    statement(s)  
else:  
    statement(s)
```

- Using Indentation to Create Blocks

Comments

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Functions

- While Syntax

```
while expression:  
    statement(s)
```

- For Syntax

```
for iterating_var in sequence:  
    statements(s)
```

- Loop Control Statements (Break, Continue)

Comments

Variables

Basic Operators

Console O/I

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While / For

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/Dictionary

Functions

- Declaration

- Accessing Values

- Updating

- Random

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**Lists / Tuples /
Dictionary**

Functions

- Declaration

```
def function_name(parameters):  
    statement(s)  
    return expression
```

- Calling

```
function_name(parameters)
```

- Default Arguments

```
def function_name(parameters = values):
```

Comments

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- Variable-length arguments

- In Python, we can pass a variable number of arguments to a function using special symbols. There are two special symbols:

- 1. *args (Non-Keyword Arguments)

- 2. **kwargs (Keyword Arguments)

```
def fun(*vars):  
    print(vars)  
  
fun(1,2,3,"hello")
```

```
def fun(**vars):  
    print(vars)  
  
fun(k1 = 1,k2 = 2,k3 = 3,k4 = "hello")
```

Comments

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- The return statement